

Customer Journey Map

Date	15 March 2026	Team ID	Not provided
Project Name	AI StudyBuddy	Maximum Marks	2 Marks

Customer Journey Map

The journey below is adapted to the supplied AI StudyBuddy backend code. The customer is treated as the student using the application; stages are based only on the implemented authentication, material/upload, Gemini processing, and stored study-content functionality shown in the code.

Phase of Journey	Stage 1 Access & Authentication	Stage 2 Upload & AI Processing	Stage 3 Study & Review
Actions What does the customer do?	- Student registers/logs in through the authentication API. - Student receives/uses a JWT access token. - Token is supplied as a Bearer token for protected requests.	- Student accesses the materials feature. - Student uploads a supported .txt, .md, or .pdf file. - Uploaded material can be processed with the Gemini integration.	- Student accesses saved material content. - Student uses generated summary, flashcards, quiz content, or study plan where provided by the materials workflow. - Student continues studying from stored material.
Touchpoint What part of the service do they interact with?	/api/auth routes - JWT Bearer authentication - protect middleware - Login/registration UI/API client	/api/materials routes - Multer upload middleware - uploads/ directory - Gemini AI service - MongoDB Material model	/api/materials responses/data - MongoDB Material document - Summary - Flashcards - Quiz - Study plan
Customer Thought What is the customer thinking?	"Can I access my study account?" "Is my request authenticated?"	"Can I upload my study file?" "Will the system process my material correctly?"	"Can I use the generated material to study?" "Are my study resources saved for later use?"
Customer Feeling What is the customer feeling?	Expecting a straightforward and secure sign-in experience.	Expecting the upload and AI processing to complete reliably.	Expecting useful, organized study resources to be available.
Process Ownership Who is in the lead on this?	Primary: Authentication/backend Components: Express auth routes, JWT middleware, User model	Primary: Materials/backend + AI integration Components: upload middleware, Materials routes, Gemini utility, MongoDB	Primary: Materials/study-content backend Components: Material model and materials routes
Opportunities How can we improve this stage?	- Provide clear authentication error messages. - Keep JWT secrets in environment variables. - Provide clear feedback for expired/invalid tokens.	- Show upload progress and validation feedback. - Handle Gemini temporary 503/429 errors gracefully. - Provide clear feedback when a file exceeds the 5 MB limit.	- Present summary, flashcards, quiz and study-plan data clearly. - Add stronger validation and user-friendly failure states. - Provide reliable persistence and retrieval of generated study content.